

Model & Product Validation Report — v2

Reverse-engineering the platform's accuracy, then applying the same validation discipline to visuals, UX and interactivity. Structured after SR 11-7 (Federal Reserve model risk guidance). Prepared September 24, 2026; trade data 756 resolved ideas, Aug 24 – Sep 23 2026; UI measured live on quantedgelabs.net including the signed-in NEXUS terminal.

Executive summary

Finding	Evidence	Status
Legacy book has negative edge — confirmed by independent re-adjudication	Replay: 24.5% hit vs 31.7% chance (z -3.6); -0.19R/idea (t -3.6)	Critical
Outcome tracker mis-scored trades	50 recorded stops never touched after publish; 26 phantom targets; 83.5% agreement	Fixed
SPX-session producer loses money	Replay -0.27R/idea, t -2.6	Fixed
'Shorts are weaker' was a tracker artifact	Replay: shorts -0.07R vs longs -0.22R	Validated
Tight stops are the #1 measured loss driver	Stop width IC +0.30 (q 1e-8); holds within each producer	Open
Measured-structure setups are the only positive signals	Base/reclaim tags +0.22R vs -0.28R without (q 0.001)	Validated
Bottom-reversal detector — edge strengthened by fix	Excess +0.82%/10d, t 4.0 (was t 3.32)	Validated
NEXUS slowness is server saturation	1 vCPU, 8 MB free, 508 MB swap; trivial API 10 s	Operator
Text contrast failed WCAG on every panel	Muted token 2.4-2.9:1 -> 4.6-5.7:1; NEXUS board now 0/77 failures	Fixed
Public pages 5-7x heavier than needed; false claims; titles 'undefined'	2.1 MB logo, 2.9 MB photo; '300+ sources', 'SOC 2 compliant'	Fixed
Dense terminal typography	732 sub-11px text uses; 74/77 board texts under 12px; 37 targets under 24px	Open

How to read this. 'R' is the trade's own risk unit: -1R is a full stop-out, +2R is twice the risk won. Every hit rate is compared with the rate a coin-flip market would produce for the same stop and target, so a wide target cannot flatter the numbers. 'Fixed' items are deployed; 'Operator' needs an account action; 'Open' is a design change recommended, not yet made.

Part A — Reverse-engineering accuracy

A1. Method

Mock tests (conceptual soundness). Production functions fed synthetic prices with known answers, including negative controls that must stay silent: 18/18 pass (v1: 15/16; the fast-V failure was fixed and its dependent tests now run).

Independent adjudication. Every resolved idea re-scored from real 5-minute bars starting the minute it was published — regular session only, gaps filled at the open, same-bar double touches counted as stops. 668 ideas, 4 without data. This is the ground truth the tracker is judged against.

Record reconstruction. 87 'duplicate_collapsed' rows removed (restatements, not ideas); expired ideas whose path crossed one barrier re-scored (45 to stop, 8 to target).

Attribution. Every conviction layer, quality tag and idea feature correlated with realized R (Spearman IC), with seeded bootstrap 95% intervals and Benjamini-Hochberg control of the false-discovery rate across ~70 simultaneous tests.

Walk-forward. Any rule chosen from data (e.g. best exit) is chosen on the first half and scored only on the second.

A2. Is the tracker telling the truth?

Tracker said	Replay: target	Replay: stop	Replay: timeout
target	34	14	12
stop	25	163	25
expired	61	193	137

Where both agree an idea resolved, they agree 83.5% of the time. The disagreements are systematic: 50 stops the replay never saw after publication — the tracker blended in the whole day's bar, so a 2 pm idea could be 'stopped' by a 10 am low. That is fixed (same-day ideas now judged on 5-minute bars from the publish minute). 'Expired' also hid resolved trades: 193 expired ideas actually hit their stop and 61 their target within the horizon.

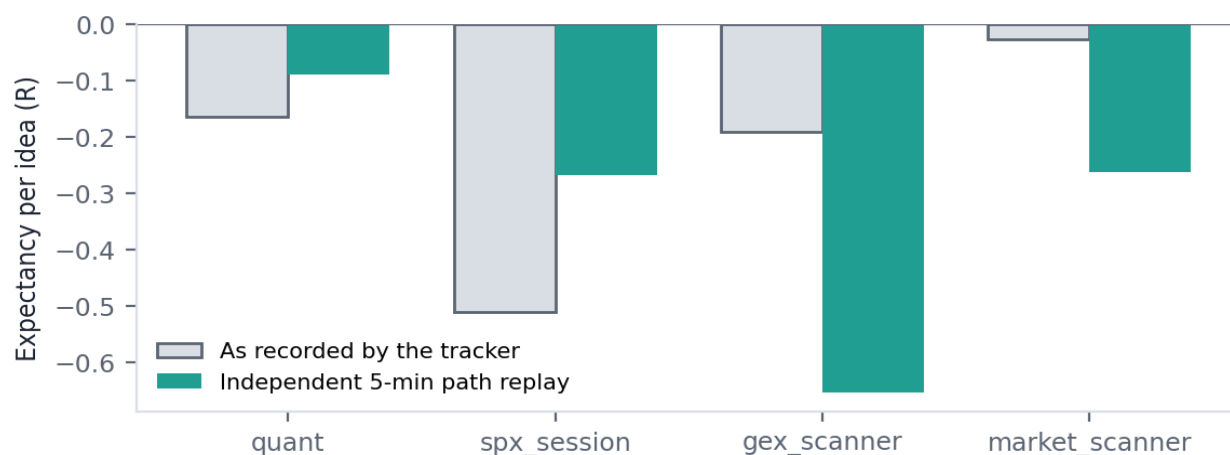


Figure 1. The tracker overstated SPX-session losses and understated market-scanner losses. The replay is the number to trust.

Book	Ideas	Hit rate	Chance	z	Expectancy	t
As recorded	756	22.3%	33.0%	-3.91	-0.24R	-4.66
Reconstructed	669	21.1%	32.1%	-4.41	-0.25R	-5.08
5-min replay (truth)	664	24.5%	31.7%	-3.57	-0.19R	-3.61
replay · long	521	21.0%	29.1%	-3.55	-0.22R	-3.81
replay · short	143	35.3%	39.9%	-1.04	-0.07R	-0.58
replay · day	256	27.5%	38.9%	-3.21	-0.21R	-2.64

Book	Ideas	Hit rate	Chance	z	Expectancy	t
replay · swing	394	23.3%	27.8%	-1.83	-0.19R	-2.75

Every lens agrees the legacy book loses $\sim 0.2R$ per idea. Shorts were not worse than longs once adjudicated honestly — the old 'shorts underperform' finding was produced by the tracker, not the market.

A3. Where accuracy is lost

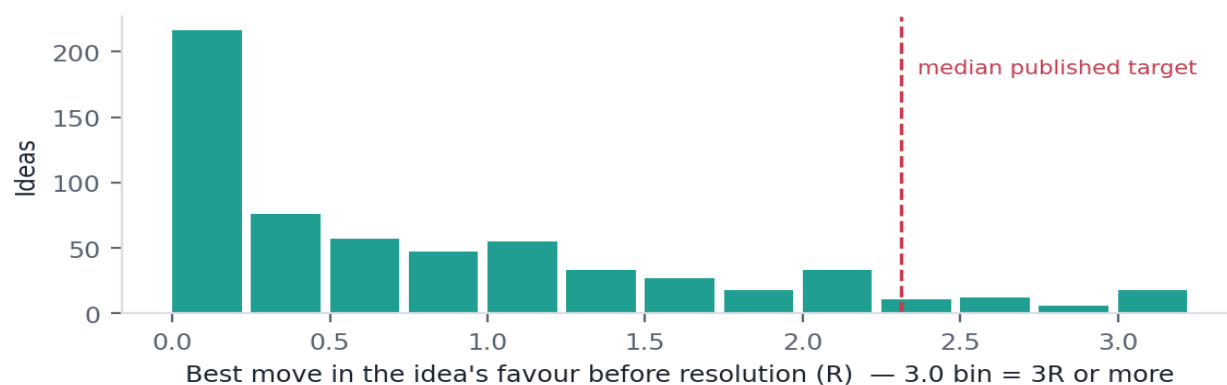


Figure 2. Half of all ideas moved at least $0.56R$ in their favour, but the median published target sat at $2.3R$. Only 35% ever reached $1R$. 55 eventual stop-outs were up a full $1R$ first.

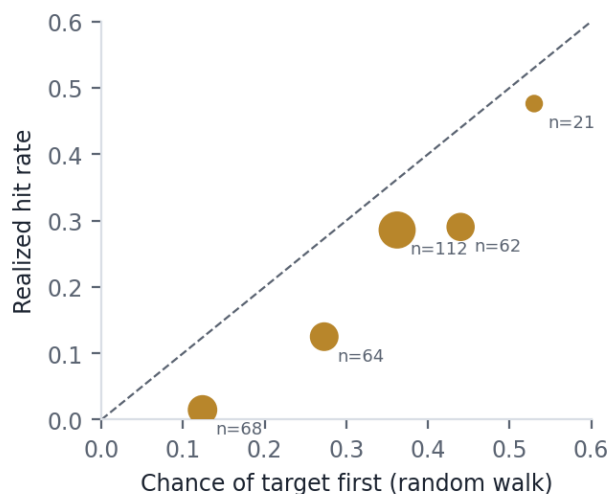


Figure 3. Realized hit rate vs the random-walk chance for the idea's own geometry, binned. Every bin sits below the diagonal: entries are selected slightly worse than random, at every target distance — the problem is not only 'targets too far'.

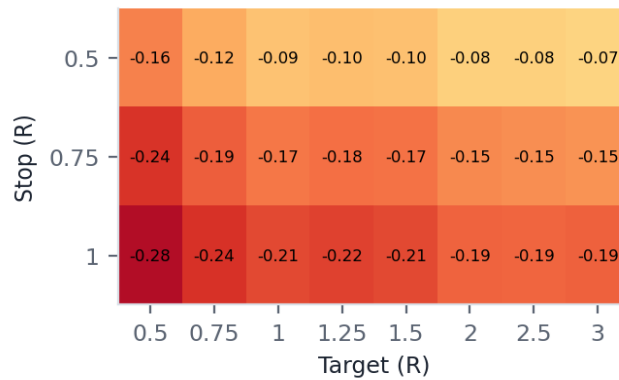


Figure 4. Counterfactual exits (pessimistic: when both levels were touched, stop assumed first). Walk-forward: the best rule on the first half (target 1R, stop 0.5R) scored -0.18R to +0.17R out of sample vs -0.27R as published. Better exits shrink the loss; they do not create an edge. Selection has to improve.

A4. What predicts a winner

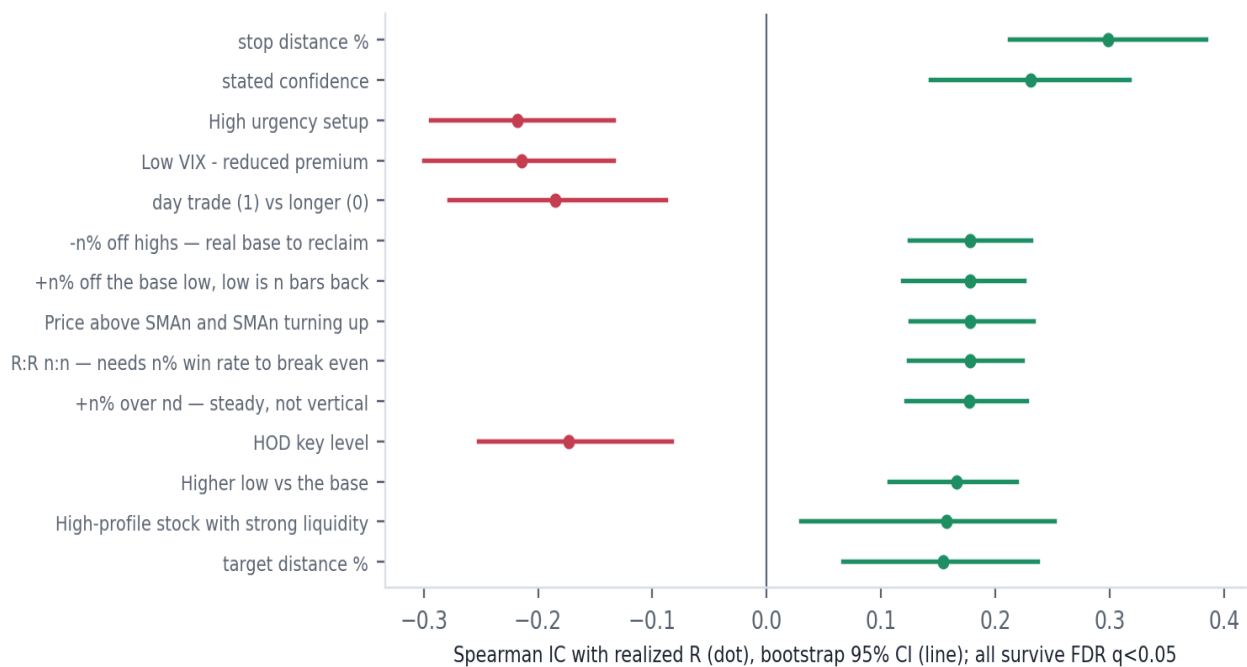


Figure 5. Features that survive false-discovery control. Green helps, red hurts.

Reading. (1) **Stop width** is the strongest single predictor: tight stops get shaken out by ordinary noise. (2) The **measured-structure tags** — real base, higher low, reclaimed average, steady (not vertical) advance — are the only positive signals, the same family as the validated reversal detector. (3) **Intraday day-trade tags** — 'High urgency', 'Low VIX', HOD/LOD key level, VWAP — are the worst. (4) **Stated confidence** ranks outcomes (IC +0.23) but is badly scaled (89% stated -> 32% realized): a usable rank, not a probability — hence its relabel to a score. (5) Of the conviction layers only **technical** carries signal; the total score does not.

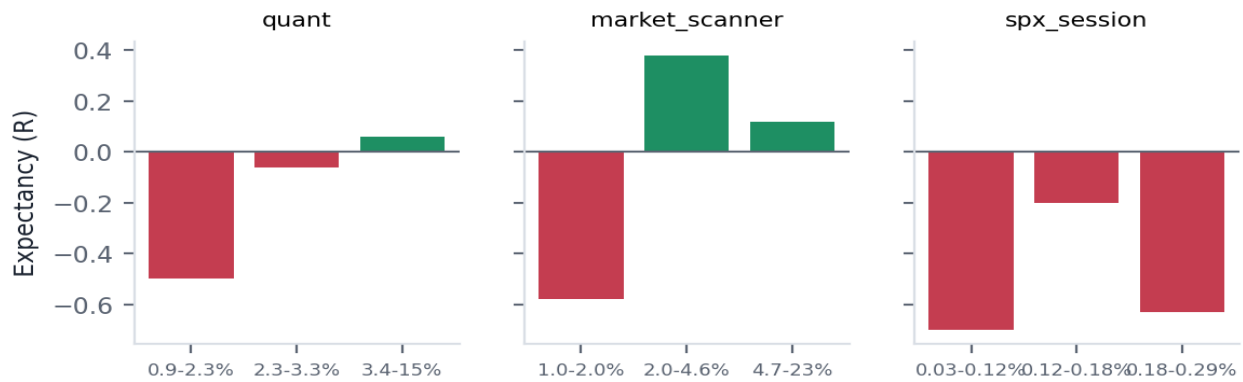


Figure 6. Control for confounding: stop width split into terciles *within* each producer. For both non-SPX producers the tightest third loses $\sim 0.5R$ and the rest break even or better — the effect is real, not an SPX artifact. Recommended: a volatility floor on stops (e.g. stop no tighter than $\sim 0.75 \times ATR(14)$ for swings).

A5. How sure are we?

Producer	n	Expectancy	Bootstrap 95% CI
quant	174	-0.14R	[-0.31, +0.04]
spx_session	151	-0.59R	[-0.71, -0.45]
market_scanner	115	+0.04R	[-0.15, +0.25]
All	459	-0.25R	[-0.35, -0.15]

Probabilistic Sharpe (Bailey & López de Prado) that the true per-trade edge is positive: $1.0e-05$ — effectively zero.

Returns are right-skewed (skew 1.5): few big winners, many small losers. **Power:** at this dispersion, confirming an edge of $0.1R$ needs ~ 878 trades; $0.2R$ needs ~ 220 ; $0.3R$ needs ~ 98 . The rebuilt scanners must accrue that many resolved ideas before any claim of edge.

A6. Detector benchmark after the fast-V fix

	Fires	Mean 10-day return	Excess vs universe	t-stat
Before (structural stop only)	2021	1.07%	+0.68%	3.32
After (ATR fallback stop)	2174	1.21%	+0.82%	4.05

Walk-forward over 37,611 stock-sessions, no look-ahead. Letting the sharpest V-recoveries qualify added 153 fires and made the edge stronger.

Part B — Visuals, UX and interactivity

B1. Same process, different model

SR 11-7 pillar	Quant (Part A)	UI / UX (Part B)
Conceptual soundness	Mock tests on production code	Static audit of all 220 screens: tokens, text sizes, semantics
Outcomes analysis	Resolved ideas, R-multiples	Runtime probe on the live site: load, layout, contrast, targets, names
Benchmarking	Random-walk break-even	Core Web Vitals (LCP 2.5 s, CLS 0.1, INP 200 ms), WCAG 2.2 AA, Apple HIG 44 pt
Stability	Chronological halves	Phone 375 / tablet 768 / desktop 1280-1440; signed-out and signed-in
Ongoing monitoring	Monthly re-run	research/ui-probe.js + ui-static-audit.py re-runnable

B2. Conceptual soundness — static audit

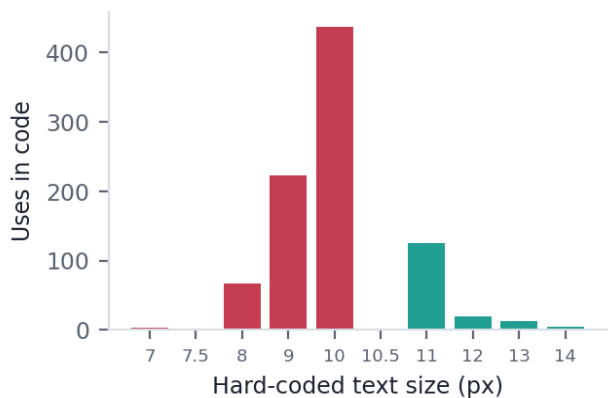
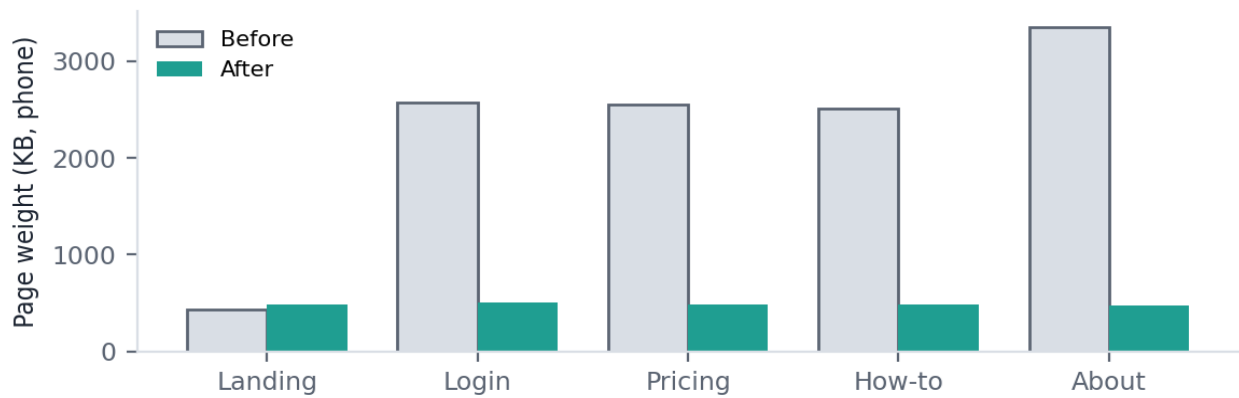


Figure 7. Hard-coded text sizes across the client. 732 uses below 11 px (red). 7-8 px text was floored at 9 px this cycle.

Measure	Value	Why it matters
Raw Tailwind palette colours	2,187 uses, 21 hues	Bypass the design tokens: theme and contrast fixes cannot reach them
Hard-coded hex in components	299 uses, 72 distinct	Same — colour drift between screens
Opacity-faded text	594 uses	Grey text at 60-70% opacity is the main contrast-failure pattern
Clickable div/span, no role/keyboards	74	Mouse-only; 8 NEXUS board rows fixed this cycle
Icon buttons without a name	31	Screen readers announce 'button'
Largest JS chunk	gex 840 KB	GEX hub ships 840 KB; split its chart/3D code
Brand marks in use	4	Blue Q, green 'QE' favicon, candlestick app icon, pulse glyph — pick one

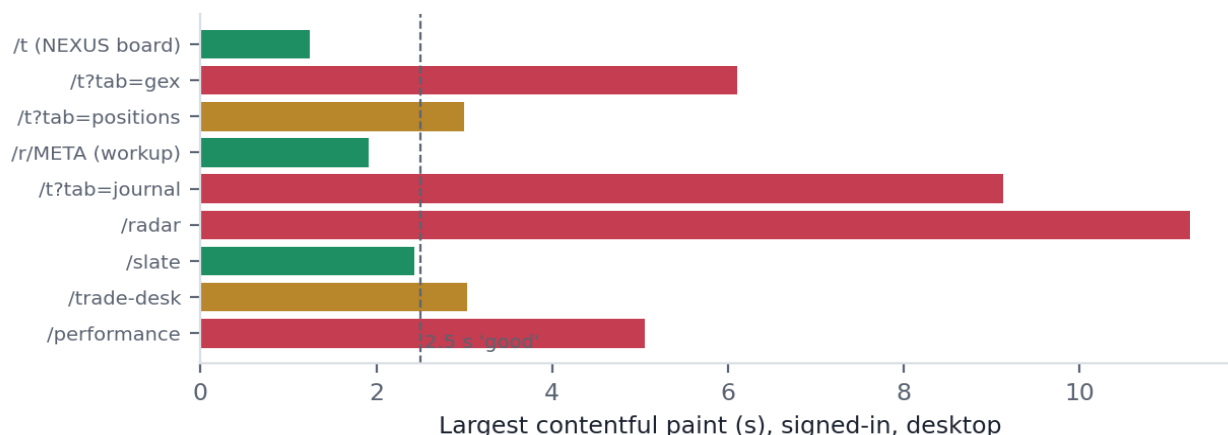
B3. Outcomes — public pages (what people you share the link with see)



Page (phone)	Weight	Contrast failures	Unlabelled inputs	Targets < 44 px
Landing	436 -> 480 KB	9/49 -> 0/47	0 -> 0	1 -> 0
Login	2,569 -> 508 KB	2/15 -> 1/13	2 -> 0	8 -> 4
Pricing	2,548 -> 486 KB	5/71 -> 5/69	0 -> 0	5 -> 4
How-to	2,507 -> 479 KB	44/109 -> 0/107	0 -> 0	13 -> 12
About	3,349 -> 475 KB	0/71 -> 0/69	0 -> 0	12 -> 11

Also fixed on public pages: every page using the SEO component shipped with the tab title 'undefined' (and blank search description); logged-out visitors saw '0 in play now' (a 401 rendered as zero) and '0 names swept'; a '34 new updates' toast greeted first-time visitors and covered the page on phones; the header logo was a 2.1 MB image that rendered as a grey smudge; the About page claimed '300+ data sources', '8000+ stocks', '6 layers' and 'SOC 2 compliant' (now: 8 feeds, 2,000 ranked names, 17 layers, SOC 2-audited cloud); the public watchlist shifted on load (CLS 0.25 'poor'); the overnight badge said 'Data stale' when the market was simply closed.

B4. Outcomes — NEXUS terminal (signed in)



Screen	LCP	Contrast fails	Text < 12 px	Targets < 24 px	API calls
/t (NEXUS board)	1.2 s	0/77	74/77	37/49	22
/t?tab=gex	6.1 s	0/1	1/1	0/12	34
/t?tab=positions	3.0 s	0/6	1/6	0/13	35
/r/META (workup)	1.9 s	12/29	27/29	8/14	36
/t?tab=journal	9.1 s	1/6	6/6	0/16	39
/radar	11.3 s	15/35	30/35	2/24	34
/slate	2.4 s	5/13	10/13	2/19	28
/trade-desk	3.0 s	5/13	10/13	2/19	37
/performance	5.1 s	5/13	10/13	2/19	48

Load forensics. A cold NEXUS board took **28.6 s** to show data; a warm load fires **78 API calls** in 20 s. The slowest were /api/market-pulse (9.2 s, uncached) and /api/extended-hours (7.5 s — and requested twice under different cache keys). Both now cache for 30 s with in-flight de-duplication: repeat calls measured 9.85 s -> 0.07 s and 8.07 s -> 0.04 s. But even trivial endpoints took ~10 s: the server has 1 vCPU at load 3.45, 8 MB free RAM and 508 MB in swap. **The 2 GB resize is the single largest UX improvement available.**

Contrast after the token fix: the NEXUS board shows 0/77 failures. Remaining failures on radar/slate/trade-desk/performance came from 9 px sidebar group labels at 30% opacity (2.2:1) and a 1.4:1 'Reading radar...' loading line — both fixed and deployed. **Density:** 74 of 77 text elements on the board are under 12 px and 37 of 49 controls are under 24 px (WCAG 2.5.8) — the terminal's deliberate Bloomberg-style density, but it costs phone and accessibility users.

Interactivity. Real clicks measured with the Event Timing API: slowest interaction 112 ms (What's-new drawer), search palette 64 ms — inside Google's 'good' INP threshold of 200 ms. Responsiveness is not the problem; data arrival is.

C. Actions taken this cycle

Area	Change	Status
Model	SPX-session publishing + its convergence vote suspended (SPX_SESSION_PUBLISH=1 to restore)	Fixed
Model	Bottom-reversal fast-V: ATR fallback stop (t 3.3 -> 4.0)	Fixed
Tracker	Same-day ideas judged on 5-min bars from the publish minute, not the whole-day bar	Fixed
Display	Signal card shows SCORE n/100, not a confidence %	Fixed
Perf	market-pulse + extended-hours: 30 s cache, in-flight dedupe, shared client key	Fixed
Perf	2.1 MB logo -> 12.5 KB; 2.9 MB photo -> 35 KB (public pages -80 to -86%)	Fixed
A11y	Muted text token to WCAG AA (dark + light); sidebar labels; radar loader; how-to	Fixed
A11y	Login labels, named toggles/links/logout, 44 px CTAs, keyboard board rows, 7-8 px text floored at 9 px	Fixed
Honesty	About claims measured; landing zeros removed; 'undefined' titles; market-closed badge	Fixed
UX	First-visit update toast removed; public watchlist skeleton (no layout shift)	Fixed

D. Recommendations

#	Action	Why	Owner
1	Resize the droplet to 2 GB (~\$12/mo)	1 vCPU in swap: 10 s for trivial reads; cold board 28.6 s	Operator
2	Volatility floor on stops (~0.75x ATR swing)	Stop width is the #1 loss driver, within every producer	Open
3	Weight conviction toward measured-structure evidence; drop intraday HOD/VWAP/urgency tags	Only structure tags carry positive, FDR-surviving signal	Open
4	Accrue ~220 resolved ideas per new scanner before claiming edge	Power analysis for a 0.2R edge	Open
5	Cut the NEXUS board fan-out (78 calls) into one aggregated board endpoint	Every call competes for one CPU	Open
6	Type scale: 11 px minimum for data, 12 px for prose; 24 px minimum targets	74/77 board texts < 12 px; 37 targets < 24 px	Open
7	Migrate raw palette colours and hex to tokens	2,187 palette uses bypass the theme	Open
8	Choose one brand mark	Four different logos in use	Operator

E. Limitations

One month, one market regime; the legacy producers dominate the resolved sample and the rebuilt scanners have too few outcomes to judge. The replay uses regular-session 5-minute bars and treats a bar touching both levels as a stop, so it is conservative. Option-premium outcomes were not replayed (delayed data; Tradier key expired). UI measurements come from one browser engine and one network location, NEXUS was measured overnight with the market closed, and static counts are pattern-based (they flag candidates, not certainties).
 Reproduce: `npx tsx research/model-validation.ts`, `research/accuracy-attribution.ts`, `research/path-replay.ts`;
`python3 research/ui-static-audit.py`; inject `research/ui-probe.js` on any page; `python3 research/build-full-report.py`.